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Construction and alignment of a high performance multipass vernier tandem Fabry-Perot interferometer

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A multipass vernier tandem Fabry-Perot interferometer is easily constructed and aligned when the two scanning mirrors are mounted on a common scanning stage. Using such a system we obtain an ultimate resolution of 10^{-3}cm^{-1} and a maximum scanning range, free of overlapping orders, of $>50\text{cm}^{-1}$. Using a 5 plus 4 pass arrangement a contrast $>10^{17}$ is obtained which, together with a high overall luminosity, permits simultaneous resolution of both weak spectral features near zero energy shift and low-frequency Raman bands. In addition to constructional details of this system we discuss the alignment procedure for uniform transmission in tandem operation, and demonstrate how the alignment requirements ultimately limit the performance of any tandem system to an "effective finesse" of <1000 . Spectra showing surface waves in silicon and shear waves in polystyrene illustrate the power of this technique.

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INTRODUCTION

It has been known for some time that the problem of overlapping orders in the Fabry-Perot interferometer may be overcome by using a vernier tandem arrangement of two etalons.¹ In this configuration two cavities of different optical spacings, d_1 and d_2 , are placed in series, adjusted to transmit simultaneously on one order, and then scanned so that at any time the change in optical cavity length Δd obeys the following condition:

$$\frac{\Delta d_1}{\Delta d_2} = \frac{d_1}{d_2}. \quad (1)$$

In this way the unique pair of coincident orders is swept through frequency space at a common rate, overcoming the order ambiguity of the conventional Fabry-Perot. The condition (1) is automatically satisfied by two pressure scanned etalons in a common tank, an arrangement used successfully by several workers.^{2,3} The useful gain in spectral range is limited in the pressure scanned tandem interferometer by the small pressure dependence of the refractive index of suitable gasses. One of us recently outlined a simple method for obtaining scan synchronization in a piezoelectrically scanned system.⁴ Two etalons are arranged in series with their axes inclined at an angle θ , and with their scanning mirrors mounted on a common stage (see Fig. 1). Thus

$$\frac{d_1}{d_2} = \frac{\Delta d_1}{\Delta d_2} = \frac{1}{\cos\theta}. \quad (2)$$

So that providing the cavity spacings are in the ratio $\cos\theta$ (which requires that both cavities have zero gap

simultaneously) condition (1) is automatically satisfied. Dil *et al.*⁵ have demonstrated a system in which scan synchronization of two separate etalons is achieved by adjusting the scanning ramp amplitudes applied to two separate etalons. This system has some potential advantage which we will discuss later. However, we consider a system with a common scanning stage easier to operate, and despite its novel cavity design, it is simple and economical to construct. Further, it does not suffer the scan linearity limitations of the separate etalon scheme. It is the purpose of this paper to amplify the original brief description of the common scanning stage interferometer.⁴ An important aspect of the use of any vernier tandem system is maintenance of scan synchronization. Very misleading results are all too easily obtained with even small departures from condition (1), and we do not feel that this aspect of operation has been adequately addressed in the literature. Although the interferometer cavity proved simple enough to construct in a college instrument shop, the characterization and mounting of the optical flats requires some care. Accordingly we arrange our discussion into the following sections: design criteria and engineering aspects; mounting and characterization of the optical flats; alignment procedures and limits on performance; and finally an outline of our overall system and its capabilities.

I. DESIGN CRITERIA AND ENGINEERING ASPECTS

The angle θ sets the ratio of the spectral ranges of the cavities and determines the overall spectral range of the tandem interferometer. If this ratio differs from unity by one part in n , then only every n th order may contribute to the tandem transmission of the system

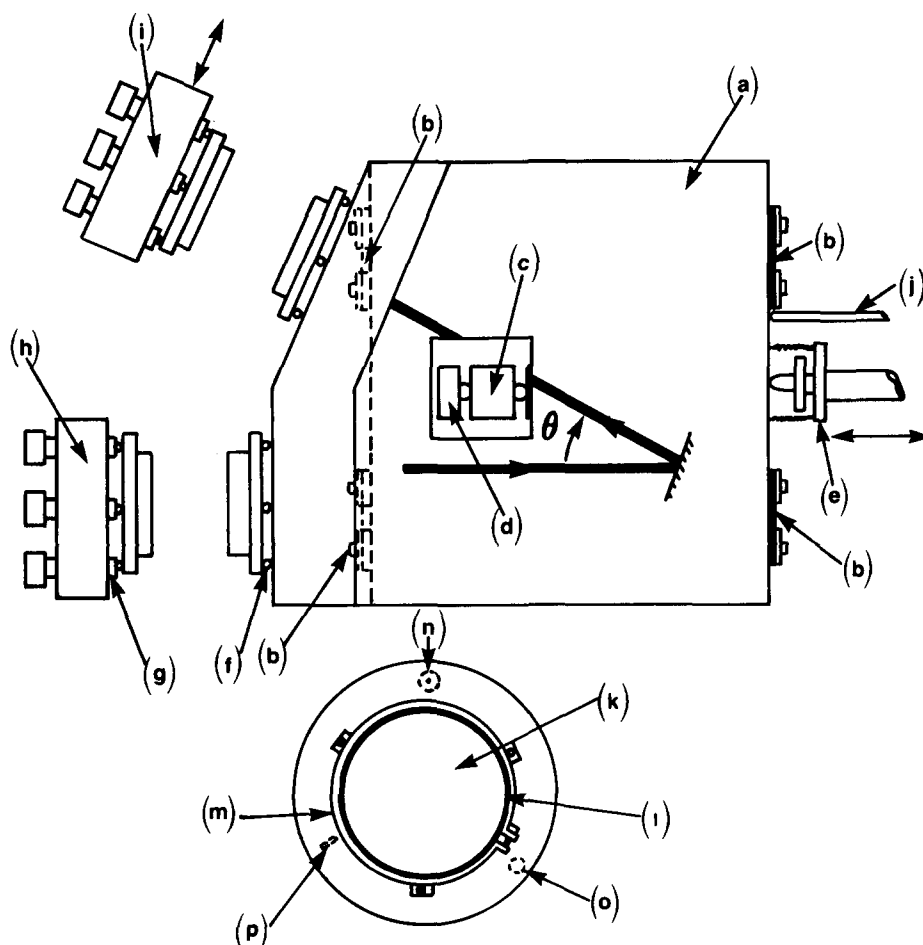


FIG. 1 The interferometer cavity. The heavy lines indicate the effective optical path through the interferometer. Labeled components are: (a) scanning stage; (b) phosphor-bronze strips; (c) PZT bimorph; (d) bearing fixed to linear bearing movement under scanning stage; (e) cavity spacing adjustment (arrow shows motion); (f) electronic plate adjustments recessed in scanning stage; (g) coarse adjustment screws; (h) fixed mirror mount; (i) translating mirror mount (arrow shows motion); (j) dial gauge tip. The detail shows a mirror mount: (k) mirror, (l) cork strip, (m) split ring, (n), (o), and (p) cone, plane, and vee bearings on reverse side of back plate.

(or near transmission, if n is not an integer). Thus the instrument's spectral range and "effective finesse" are increased by a factor n . (By "effective finesse" we mean the ratio of the accessible scan range to the instrumental linewidth.) On the other hand the transmission of a spectral feature which is off resonance by $1/n$ th of an order should be very low to suppress ghosts which arise from partial overlap of orders neighboring the coincident pair. Ghosts which arise from the elastically scattered peak cause no difficulty in interpretation of spectra because of their small width and predictable frequencies. More annoying are Brillouin ghosts derived from strong spectral features; n should be small enough to suppress these features to well below the dark count in most experiments. Assuming two 5 passed etalons each with a multipass finesse ~ 70 , computer calculations of first ghost intensity against scanning range indicate an optimum ratio d_2/d_1 in the range 0.92–0.94, or $\theta \sim 23^\circ$ – 20° . This yields a first ghost suppression of $\sim 10^5$ and a scanning range of ~ 20 orders. Of course, if scanning range is not important, as for example when the second etalon is a confocal cavity used for high resolution linewidth measurements, then a much larger value of θ may be used to some advantage.

The second aspect which merits some consideration is the number of passes used in each etalon. We have opted for the apparently complex arrangement of 5

passes in the first etalon and 4 in the second. This approach is not founded entirely on masochism, for the following advantages arise: (a) Enhanced servo control; (b) Very high contrast with consequent ghost suppression; and (c) A very serviceable conventional multipass etalon easily switched into operation (as described later) when quick measurements of simple spectra are required. For this function 5 passes are about optimal.⁶

With suitable attention to the use of the optical flats the extra passes cause negligible loss of transmission and make maximum use of the required two pairs of flats. More than two passes enhance the servo operation for two reasons. First, when one or two passes are used in the second etalon, tilts can give rise to an overall change in the relative cavity spacing, ΔZ . Since the tandem arrangement is most sensitive to ΔZ adjustment, the servo can find local maxima in the overall transmission for which ΔZ and tilt adjustments compensate each other on alternate servo cycles. This interaction can be avoided if the aperture(s) are symmetrically placed about the center of the tilt axes; however, such a symmetry constraint may not be compatible with use of the least aberrant areas of the mirrors. The second reason is that three or more non-collinear passes define a plane, causing inherently greater sensitivity of the multipass transmission.⁶

The mechanical arrangement of the interferometer cavity is shown in Fig. 1. The scanning stage consists of a Meehanite casting which overhangs a linear bearing table, and is connected to it by four phosphor-bronze strips to form a double parallelogram which is scanned linearly by a single PZT pusher resting between the Meehanite casting and a fixed bearing on the linear bearing table top. The use of such parallelogram deflection arrangements has been extensively discussed by Jones,^{7,8} and the only critical step lies in ensuring that the long sides of each parallelogram are equal to within $\sim 10\mu\text{m}$. This is easily achieved by simultaneous milling of the linear bearing top and Meehanite casting. Using a bimorph fabricated from Philips PXE-4 material, we are able to scan >20 orders for $\pm 200\text{ V}$ applied with no detectable loss of transmission arising from twist. Although the mass of the Meehanite casting renders it insensitive to placement of optical components on the scanning stage (greatly facilitating the setup of the system), it does have a low resonant frequency, and a slew-rate limiter is used on flyback to eliminate mechanical ringing. The linear bearing table is itself mounted to a Meehanite base onto which are bolted the two fixed mirror mounts, also Meehanite castings. The linear bearing (Schneeberger NK100) is good enough to allow changes in the spectral range of several centimeters with little loss of alignment—usually easily recoverable on the electronic PZT adjustments. Changing the spectral range thus becomes a trivial operation. The linear bearing table is simply slid to the desired gap by unlocking a rod which is spring loaded against it. (The spring loaded flexible bearing ensures that the linear bearing is not twisted on relocking.) A screw bearing in the rod provides fine gap adjustment whilst the cavity spacing can be read directly from a dial gauge to $\pm 5\mu\text{m}$. Large changes of cavity spacing produce a loss of aperture, unless the second etalon and its associated optics are translated sideways to recombine the apertures. We provide for this by mounting the second etalon fixed mirror and its associated optics on a sliding dovetail which translates on an axis perpendicular to the cavity axis. The mount on which the mirror is held must be firm and securely locked when in operation, though the mounting of the associated corner cubes is not critical. (The relatively small value of θ used means that this realignment is only necessary when very large changes are made in the cavity spacing, but the relative cavity spacing should be checked after such adjustments.)

An important consequence of the use of a single scanning stage is that nonlinearities in the single PZT pusher do not affect tandem transmission. In an arrangement using separate etalons⁵ nonlinearity in the PZT scanners ultimately limits performance, a point we shall address. Actually the single pusher makes servo control of linearity remarkably easy: a parallel plate capacitor of $\sim 100\text{ pF}$ is mounted with one plate scanning, the other fixed to the linear bearing table as described elsewhere.⁴ The capacitor forms part of an ac bridge

whose output is proportional to the gap; it is used in conjunction with the scanning ramp to derive an error signal which is applied as a correction to the high voltage driver that scans the stage. The servo is easily adjusted to obtain a scan of over 20 orders with no detectable nonlinearity recorded in a 1024-channel scan.

II. MOUNTING AND CHARACTERIZATION OF THE OPTICAL FLATS

The optical flats are the most important part of any etalon system, and although Dil has discussed this matter in some detail,⁵ it is worth reemphasizing the point that theoretical finesse and contrast can be obtained if care is taken in selecting and mounting mirrors. Whilst concurring with Dil in concluding that the external cavity formed by the mirror-reflector pair does not degrade performance, we would add to his arguments the point that Michelson quality corner cubes should be avoided—the coherence scrambling provided by lower quality components of sufficient angular accuracy helps to spoil this cavity's Q .⁶ The key to selecting and mounting optical flats lies in making accurate interferograms. Not only does this enable the selection of good flats (we have examined one commercially available set of " $\lambda/200$ " plates which contained significant regions out by $\lambda/10!$), but the vibration susceptibility and long-term stability of the interferometer are best measured at this point. To make an interferogram we use the following procedure: A well collimated beam of monochromatic light floods the plates at normal incidence (a shearing plate is an asset here). However, instead of examining the transmitted beam, we use the expedient of every optical workshop—a high quality beam splitter is placed at 45° in the incident beam, picking off the back reflection from the etalon. This back reflection is used to record the interferogram. The cavity is now aligned so as to minimize this reflection. Departure from perfect flatness is evidenced by bright areas in the interferogram. Of course the contrast is enormous compared with the practice of recording transmission interferograms, and effects due to misalignment or plate mounting stress are very easily separated from the inherent plate figure. With the scanning PZT connected to a high quality power supply and the plates carefully aligned, contours of equal separation may now be mapped by advancing the scanning stage, previously calibrated by recording the $\lambda/2$ voltage. In this manner we can see definite changes in the interferogram corresponding to a $3\text{-}\text{\AA}$ advance of the scanning plate (the lateral resolution is limited to $\sim \lambda$, of course), and in Fig. 2 we show a typical interferogram recorded in this case with flats which were better than $\lambda/200$. The ability to record such interferograms is a prerequisite to good interferometer operation. Our interferograms indicate that much of the aperture we use is good to $\sim \lambda/300$, in that laterally extensive aberrations (such as the $\lambda/200$ "island" at the top of the interferogram in Fig. 2) are avoided. The lack of lateral resolu-

tion hides rapid local variations in plate flatness and it is these unresolved variations which ultimately limit the performance of our system. An accurate interferogram indicates adjustments that need to be made in plate mounting and, after such adjustments, which areas of the plates should be used.

Our optical flats were supplied by ICOS of London, soft coated (ZnS-MgF_2) for a reflectivity of 91% and matched to better than $\lambda/200$ in the green over most of their aperture of 50 mm. (ICOS will supply mirrors with a plate figure specified *after* coating). We obtain a single pass finesse approaching 30, which results in a multi-pass finesse of >60 . To utilize flats of such quality the mirror mountings must be essentially stress free. Dil achieves such mounting with free horizontal supports,⁵ a technique not easily applied to a tilted etalon pair. Following an earlier approach of Dil's⁹ we mount the flats in split rings fabricated from maraging steel (see the detail in Fig. 1), using a cork strip to relieve local stress points. The tension in the ring is adjusted with the aid of interferograms, and in practice we find a fair range of adjustment between the point at which the mirrors are effectively gripped and the point at which detectable bowing occurs. It is important that the mounting rings themselves are not stressed by adjustment, so we spring load the rings against a set of kinematic jeweled bearings (a plane, cone, and vee) arranged in the optimum 120° geometry.⁶ Coarse adjustment is made with direct-drive micrometers on the fixed mirror mounts. The direct drive is more stable than a differential arrangement, and the $100\text{ }\mu\text{m}$ thread pitch we use is cut into a 0.5-in. Invar rod using a Hardinge precision lathe. Dry lubricants are used in order to avoid creep. The mirrors on the scanning stage are also kinematically mounted onto a 120° arrangement of PZT transducers. Our transducers consist of six 0.75-in. diam discs of Vernitron PZT 5-H material driven in parallel. Although less sensitive than a bimorph, the stacked arrangement has a much smaller bending moment and is less susceptible to vibration. With a control voltage in the range 0–600 V the electronic adjustment is adequate to eliminate the need for mechanical adjustment in normal operation. The 120° mounting geometry requires that the electronic drive be arranged so as to give orthogonal tilt control,⁶ in this case including separation of tilt and axial spacing adjustments. Once the mirrors are mounted, characterized, and vibration free, the optimum location for each pass may be selected. Since the aberrations may not be circular, greatest flexibility is achieved by using one corner cube reflector for each pass, cut so as to allow neighboring beams to pass, and mounted to allow X–Y translation and rotation of the reflectors, the positioning of which is not critical. (Stray reflections are minimized by some tilt of the corner cube faces and by using reasonable optical paths consistent with the overall diffraction limited optical path requirement.) Our 9 passes overall require 7 retroreflectors, a number which imposes budget constraints on corner cube quality—in our experience the

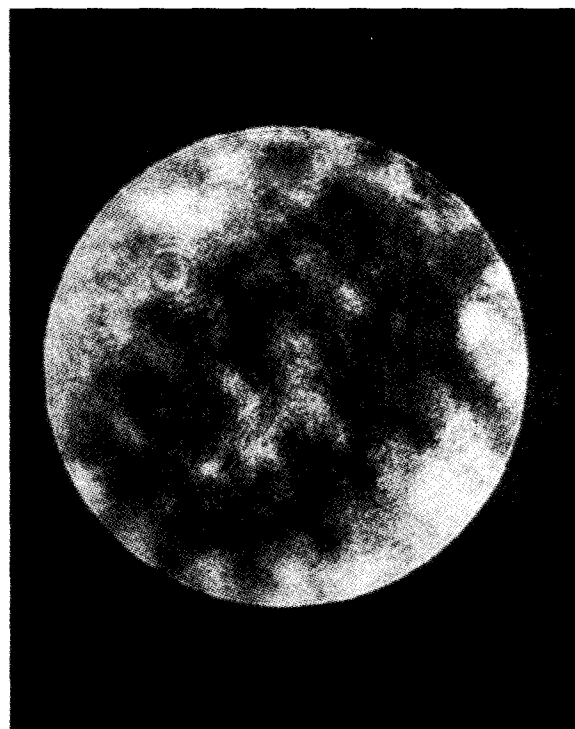


FIG. 2. A back-reflection interferogram of one pair of etalon mirrors (recorded on Polaroid type 55 film). The entire 50 mm aperture is shown. The axial movement required to darken the brightest features is 2.2 nm, indicating that these mirrors are matched to better than $\lambda/200$. Changes are seen in the fine detail for cavity movements as small as 3 Å. The bright region on the top left of the interferogram is the $\lambda/200$ "island" mentioned in the text.

cost of these components is inversely related to the area of chamfer at the cube corners, a manufacturing practice which accounts for a significant fraction of our losses! Chamfer losses aside we have a clear aperture of 16 mm in 9 pass operation, which together with the high overall transmission obtained with good optical flats gives us adequate luminosity to collect Raman spectra from such difficult samples as metals or water in a reasonable time.

III. ALIGNMENT PROCEDURES AND LIMITS ON PERFORMANCE

In operation each etalon is first brought into alignment using monochromatic light, then one order, n , of the first etalon (gap d_1) arranged to overlap one order, m , of the second (a " ΔZ " adjustment is obtained by adding a control voltage to all 3 stacks on the second etalon). If when d_1 is zero, the second etalon has a gap ϵ , then after orders are made to coincide

$$n\lambda = 2d_1,$$

$$m\lambda = 2d_1 \cos\theta + \epsilon. \quad (3)$$

When the scanning stage is displaced a distance Δ , the ratio of the wavelengths transmitted by each cavity simultaneously, λ_2/λ_1 , is

$$\frac{\lambda_2}{\lambda_1} \approx 1 - \frac{\Delta\epsilon}{[(d_1 + \Delta) \cos\theta + \epsilon]d_1}. \quad (4)$$

In practice the relative misalignment should be much less than the instrumental linewidth, $\delta\lambda$. For an etalon of operating finesse $F \gg 1$, this quantity is given by

$$\delta\lambda = \frac{\lambda^2}{2dF}, \quad (5)$$

so that a 75% loss in the transmission of the tandem arrangement occurs when $\epsilon = \epsilon_F$, given by

$$\epsilon_F = \frac{d_1 \cos\theta}{NF}, \quad (6)$$

where N is the number of orders scanned. Thus scanning ± 10 orders each corresponding to 2 cm^{-1} with a finesse of 60 requires $\epsilon \ll 4 \text{ }\mu\text{m}$! At this point we must stress the difficulties inherent in using a narrow spectral feature as a guide to alignment.⁵ With gross misalignment the relative movement of transmission profiles may be enough to transfer transmission to neighboring orders ($n \pm 1$, $m \pm 1$) within the scan range. In the example just given this would require slipping 1/20th of an order (for $\cos\theta = 20^\circ$) in 10 orders or an error, $\epsilon \sim 12 \text{ }\mu\text{m}$. In an arrangement where two separate etalons are synchronized by adjusting the scanning ramp amplitudes,⁵ it is clearly insufficient merely to adjust the ramp ratios so as to maximize some Brillouin component—in cases where the gap ratio is near to unity, very little error is sufficient to bring adjacent orders into alignment on a Brillouin peak. Clearly the required scan ratio must be calculated from the measured gap ratio with sufficient accuracy, and alignment completed by recording an appropriately broad spectral feature. Our reluctance to use the two-etalon approach stems from this difficulty, since this procedure would have to be repeated whenever the etalon gaps were changed. Another difficulty lies in the insensitivity of

features near the center of the scan to small errors in relative spacing. Both these difficulties are illustrated in Fig. 3 where we display the $V-H$ and $V-V$ spectrum of liquid nitrobenzene recorded with a deliberate misalignment ($d_1 = 2.70 \text{ mm}$ and $\epsilon = 350 \text{ }\mu\text{m}$). The $V-H$ spectrum consists of two broad Lorentzian components quite intense out to $\pm 10 \text{ cm}^{-1}$.¹⁰ However, the transfer of transmission to neighboring orders every half order scanned results in the distorted spectrum shown in Fig. 3. Full transmission occurs at each local maximum so there are many incorrect adjustments of the plate spacings which can produce a maximum in the transmission of sharp Brillouin peaks. The second difficulty stems from the relatively flat nature of the multipass profile near its center (see, for example, the transmission vs misalignment curves for the multipass Fabry-Perot in Ref. 6). Maximizing the transmission of Brillouin peaks near the center of a scan is not a sensitive test of transmission at the extremes of the scan. This is illustrated rather dramatically by Fig. 3 also. The fully polarized spectrum (shown on a reduced scale at the center of the spectrum) contains LA components at $\pm 0.2 \text{ cm}^{-1}$. This peak intensity is hardly affected by even this gross misalignment.

Interferometric methods may be used to measure the gaps and in principle set $\epsilon = 0$. For example, the plates may be flooded with a convergent monochromatic beam to produce a ring system on a screen. The distance moved to go from one collapsed central order plus one ring of a given angular diameter to a collapsed central order plus a second order ring of the same angular diameter is clearly twice the original plate spacing. We have found this method laborious in that the multipassing optics and screens must be removed, and it is limited by the accuracy of the dial gauge and the fringe widths to measuring d to $\pm 5 \text{ }\mu\text{m}$ corresponding to a possible ϵ of $\pm 10 \text{ }\mu\text{m}$. We find the easiest approach is to set one pair of plates to a small gap (an end stop prevents us from bringing the plates closer than $\sim 100 \text{ }\mu\text{m}$, a good size for the ever present Arizona dust), and then to adjust the second gap by eye to a similar value. The process is carried out with the plates illuminated with an alignment beam so that accidental contacting of the flats is avoided by maintaining reasonable parallelism during the final stages of adjustment. With d_1 set at 2 or 3 mm we then record the depolarized spectrum of liquid nitrobenzene, adjusting one gap so as to obtain a broad central feature with smooth tails quite intense out to 10 cm^{-1} or so. The alignment is then completed by placing an incandescent lamp in the sample chamber (allowing a little laser light to hit the filament, so as to provide a sharp feature for servo control) and adjusting this time for a flat background. Whilst this is easier than attempting to obtain a precise fit to the nitrobenzene depolarized spectrum, it is a step that should be taken only after the machine is very nearly in alignment already. This is because the white light spectrum contains contributions from overlap (or near overlap) of far removed orders. It is possible to obtain

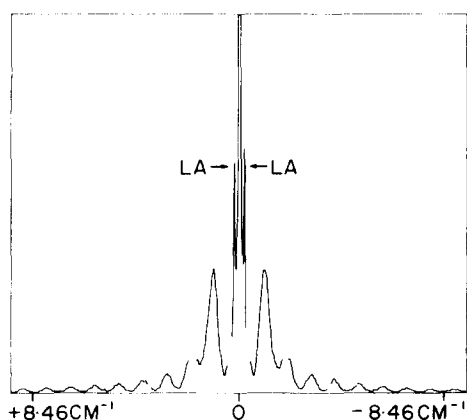


FIG. 3. Spectrum of liquid nitrobenzene obtained with a grossly misaligned instrument ($d_1 = 2.70 \text{ mm}$ and $\epsilon = 350 \text{ }\mu\text{m}$). The polarized ($V-V$) spectrum is shown at the center, while beyond $\pm 0.4 \text{ cm}^{-1}$ the depolarized ($V-H$) spectrum is shown. The broad depolarized scattering¹⁰ is distorted by the order to order slipping caused by the misalignment. The L.A. phonon intensity (peaks at $\pm 0.2 \text{ cm}^{-1}$) is little affected by misalignment, as would be the case for sharp features near subsequent maxima in the transmission. On this and subsequent spectra, connected raw data points are shown, and ghosts are removed for clarity.

a background consisting of many adjacent peaks which correspond to order-to-order slipping of many sequences of far removed orders. Such a background can look deceptively flat. Of course, this operation requires the use of an additional broadband filter. A diffraction grating is incorporated into our optical train, chosen so as to give a passband somewhat greater than the total scan of the machine when used with an angular aperture appropriate to the desired Fabry–Perot resolution.¹¹ At the expense of some luminosity this aperture may be further reduced to prevent spurious signals in the rare event that the sample has a strong Raman band at ~ 20 orders shift.

The requirement that any alignment error be significantly less than the quantity ϵ_F given by Eq. (6) ultimately limits the performance of the machine: even if the relative thermal drift in the cavity spacing is negligible, it may still be necessary to vary the gap of one etalon over 1 order or $0.25\ \mu\text{m}$ to obtain the desired overlap. This means that scanning ± 10 orders with a finesse of 60 would lead to severe distortion of the spectrum for gaps smaller than $\sim 150\ \mu\text{m}$. Whilst this corresponds to a $\sim 300\ \text{cm}^{-1}$ scanning range with a resolution of $0.6\ \text{cm}^{-1}$, it gives a rather limited access to both the conventional Raman and Brillouin frequency regimes. It might be argued that the separate etalon technique overcomes this difficulty since one has the freedom to adjust the scanning ramp amplitude to whatever ratio is necessary, and herein lies the possible advantage of this scheme. However, unlike the present arrangement, a severe restriction is imposed by the nonlinearities of the PZT stacks when separate etalons are used. It is of course possible to program the scanning ramps to reduce this effect.¹² The current “state of the art” claimed for a commercial system is one part in 250 scanning 6 orders.⁵ It is not clear whether this is a point to point or integrated departure from linearity; however, unless the characteristics of the separate etalons are identical, this nonlinearity could give rise to a relative displacement of the two transmission peaks by as much as $1/125$ th of the total scan range, limiting the “effective finesse” to just 125. This is because each etalon must be limited to a single etalon multipass finesse of 20, so that on scanning six orders the previously coincident pair of orders is misaligned by about an instrumental full width, causing a 75% loss of transmission.

IV. LAYOUT AND PERFORMANCE OF THE SPECTROMETER

A schematic layout of the spectrometer we use is shown in Fig. 4. The spatial filter SF1 is used to define the angular acceptance of the interferometer, whilst SF2 serves to define the resolution of the passband filter which utilizes a blazed grating Gr in a near Littrow configuration. (SF2 is limited at the very largest passbands by the need to reject spurious reflections from the mirror wedges at 0.25° off axis.) C5 is a retro-reflector which may be moved out of position to permit

operation of the 5 pass etalon alone. In position the beam is deflected back onto the mirror M3 to enter the 4 pass etalon for tandem operation. Since the output of the second etalon is picked off by M3 also, the tandem and 5 pass output share a common axis. The aperture of the subsequent optical system is sufficient to allow transfer between 5 pass and tandem operation merely by switching in C5 which is of sufficient accuracy to render realignment of SF2 unnecessary. It is of course necessary to prealign the second etalon before attempting tandem operation, and preferable to align both without the need to remove a sample. Consequently, a little of the laser light is picked off with beam splitter B1, fed into the spectrometer box past the shutter S1, expanded and filtered in SF3, and fed into each etalon via the beam splitter B2 and the alignment mirrors AM1, 2. AM3 and AM4 pick off the transmitted beams which are directed to photodiodes PD1 and PD2 for display separately, or as a summed signal to aid tandem alignment. The mirrors AM1–4 and shutter S1 are switched into position by a common signal which also throws a protective mirror, S2, in front of the ITT FW130 detector. S2 also serves to provide a view of the final pinhole for alignment purposes. A weak high order reflection from the grating, Gr, is used to drive the servo detector PM. The servo is an appropriately extended version of the system described elsewhere.¹³ Analog gates select which one of 5 axes, X_1 , Y_1 , X_2 , Y_2 (parallelism), or ΔZ (relative cavity spacing) is controlled. The shutter S3 is sent into the beam at a preset count rate to protect the FW130 from strong Rayleigh signals. It is withdrawn when the servo tube signal falls below the level sampled at the point of shutter insertion. The mirrors AM5 and AM6 may be switched into the optical path, providing either a view of the pinhole in SF1, or allowing a helium–neon laser beam to be sent through SF1, facilitating sample alignment.

The instrument has proved versatile and particularly easy to operate. Changing the spectral range of the tandem arrangement is an easier operation than we have experienced in working with any commercial conventional single pass etalon. Set-up time is minimal, and the machine holds servo lock indefinitely, although we do provide for unattended operation by arranging a shutdown of the signal averager, if the servo signal falls below a preset level. We tabulate in Table I the theoretical and measured performance of the spectrometer, although we should point out that it is only possible to set a lower limit on the contrast, whilst transmission measurements are difficult to make with any certainty. A qualitative indication of the contrast is given by the rather hazardous procedure of running the laser directly into the machine. Off resonance *no* light is detectable. (The grating rejects plasma lines, of course.)

We believe the difference between the observed and calculated finesse is due to the laterally unresolved mirror aberrations (i.e., rapid local variations in surface quality of lateral extent $< \lambda$). An indication of the

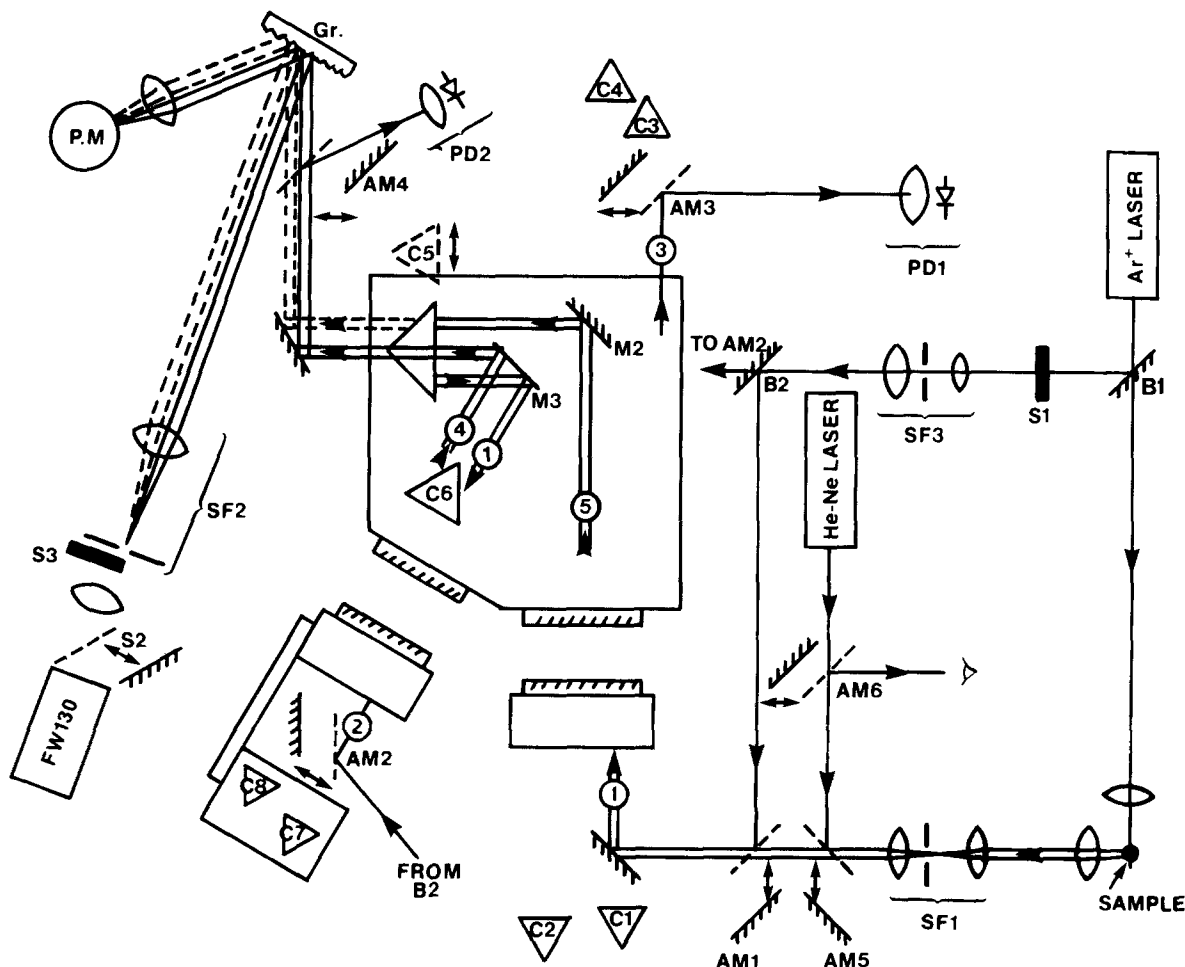


FIG. 4. Overall schematic layout of the spectrometer. A single heavy line shows the path of alignment beams. The double line shows the path of signal beams (the dashed double line indicating the continuation of the 5 pass output with C5 switched out). For clarity we omit beam paths within each etalon system. The retroreflectors C1, 2, 3, and 4 form the multipassing optics for the first etalon. Multipassing of the second etalon is achieved with C6, 7, and 8. Numbers within circles indicate the destination or source pass of beams. Components which are selectively switched in are shown dashed in the beam path with the associated double headed arrows indicating the movement.

existence of these aberrations is the rapid change in the very faint speckle pattern observable in the reflection interferograms as very small ($\sim 3 \text{ \AA}$) adjustments are made to the cavity spacing.

Finally, in Fig. 5 we show two spectra recorded with this interferometer. Figure 5(a) is a spectrum taken in internal backscattering near the $[111]$ axis of a boron

doped silicon crystal. The external angle to the $[111]$ direction was 76° , allowing surface waves on the (111) plane to be observed [labeled 'R' in Fig. 5(a)]. These measurements are similar to those reported elsewhere,¹⁴ although the resolution of two TA components should be noted. The ability of the instrument to detect weak features close to the Rayleigh line is illustrated in Fig. 5(b) which shows both $V-V$ and $V-H$ scattering from polystyrene at 90° . The features labeled TA are fully depolarized in the 90° geometry; their Brillouin shift corresponds to the frequency calculated for shear waves from the low frequency values of Young's modulus and Poisson's ratio, and the LA/TA intensity is in line with the known ratio of the Pockels coefficients P_{12}/P_{44} .¹⁵ We conclude that these peaks arise from hypersonic shear wave propagation. We note also the broad depolarized central feature which is not instrumental in origin. (The slight asymmetry of the instrumental peak is caused by heating of the FW130 photocathode.) It had been argued that, with the exception of polycarbonate, hypersonic shear waves would be unobservable in polymers because of their low intensity.¹⁵

TABLE I. Summary of the performance of the spectrometer.

	Calculated	Observed
Transmission through tandem etalon pair (full 16 mm aperture)	0.30	≥ 0.10
Contrast	4×10^{25}	$\geq 10^{17}$
Finesse (conventional etalon definition)	78	60
Scanning range possible without loss of transmission (20 orders, i.e., ± 10 , scanned)	$\pm 0.55 \text{ cm}^{-1}$ (10^{-3} cm^{-1} resolution) to $\pm 50 \text{ cm}^{-1}$ (0.08 cm^{-1} resolution) ^a	

^a This corresponds to the minimum gap for which proper tandem alignment is easily attained.

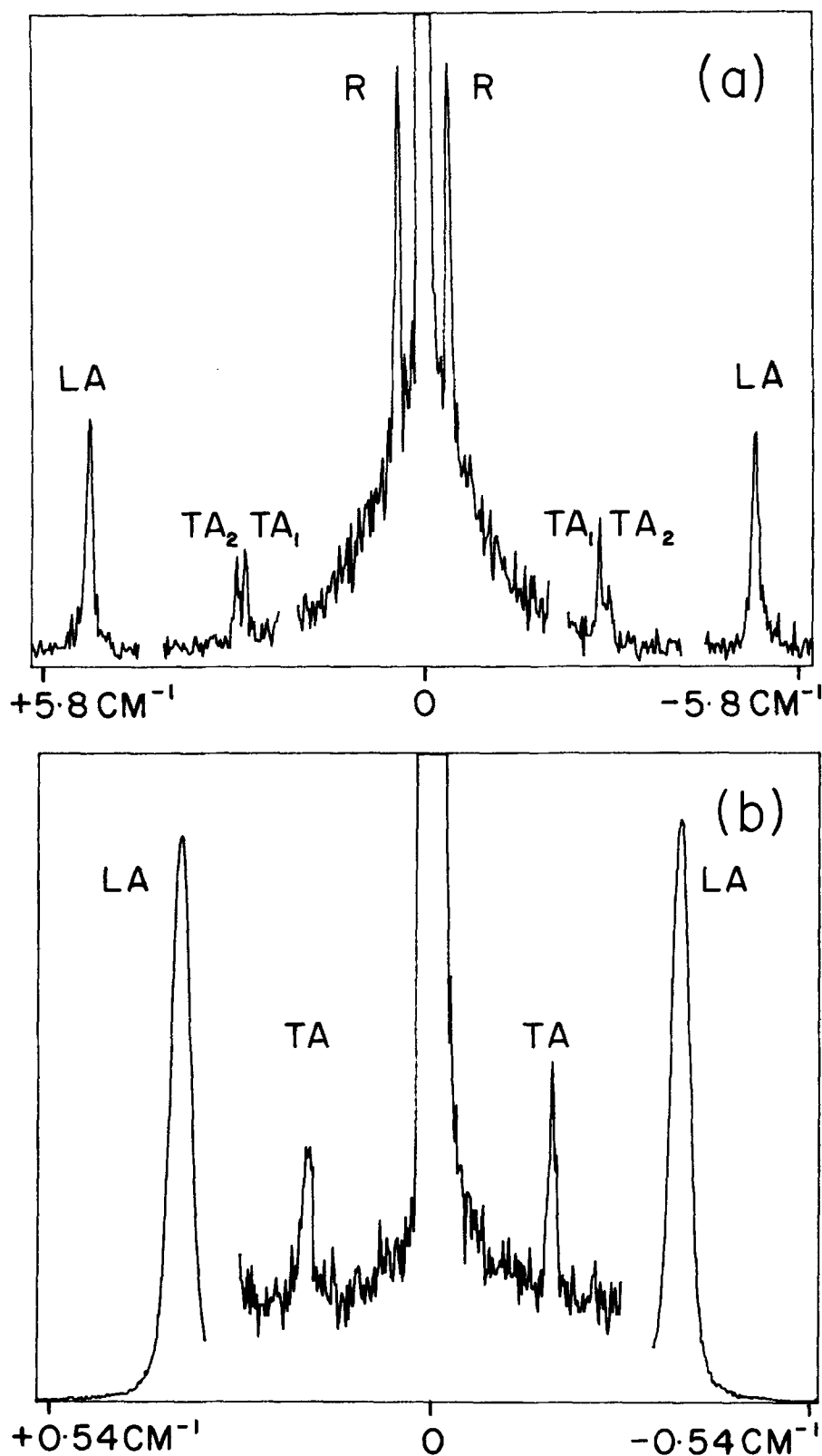


FIG. 5. (a) Unresolved (V+H) spectrum of boron doped silicon. The spectrum was obtained in internal backscattering with the light incident at Brewster's angle on the (111) plane. The surface waves are labeled R. The internal scattering occurs at a small angle to the [111] direction and some splitting of the TA peaks is resolved. The central feature is more pronounced than that observed in earlier work.¹⁴ (b) Spectrum obtained from polystyrene at 90° scattering angle. The portion of the spectrum beyond $\pm 0.28 \text{ cm}^{-1}$ is fully polarized (V-V) and reduced in scale by a factor of approximately 200 relative to the central portion which is fully depolarized (V-H).

V. DISCUSSION

The vernier tandem interferometer we describe is extremely easy to use, and its design and construction are straightforward. The advantages of a 5 plus 4 pass configuration may be realized at little cost in transmission, providing adequate care is taken in selecting,

mounting, and utilizing the optical flats. The system must be carefully aligned for correct tandem operation, and we outline a procedure suitable for adjusting the plate gaps in our system, or selecting the scanning ramp ratio when separate etalons are used. The system we have constructed offers a greatly extended spectral range with high resolution and very great contrast. The con-

trast is sufficient to allow resolution of a spectral feature displaced by the instrumental resolution from the Rayleigh line, even where the feature intensity itself is comparable to the FW 130 dark count (<0.5 cts/s), and the Rayleigh line arises from an opaque or semi-opaque material such as silicon. Thus we very nearly bridge the gap between correlation and Raman spectroscopy, and do so without constraint on the nature of the sample studied.

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trum of polystyrene [J. K. Krüger, L. Peetz, W. Wildner, and M. Pietralla, *Polymer* **21**, 620 (1980)]. The polarization analysis in this work supports the original assignment of these peaks by Krüger *et al.*, although the intensity ratio (TA/LA) they report is not consistent with the ratio calculated from the known Pockels coefficients.

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